A Classical AI Reasoning System for Logical Entailment

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1. Overview and Internal Representation

This report describes a classical symbolic reasoning system that determines whether a query Q follows from a structured knowledge base, i.e., whether $KB \models Q$. Its output is binary: `True` or `False`. The intended interface is

$$(\text{facts}, \text{rules}, \text{query}) \longrightarrow \text{reasoner} \longrightarrow \{\texttt{True}, \texttt{False}\}.$$

The implementation uses symbolic procedures rather than an LLM or a pretrained neural model. It operates on structured facts and rules, formal FOL annotations, or causal mechanisms supplied by dataset adapters; it does not attempt unrestricted natural-language understanding.

The engine uses three principal representations:

- **Ground and pattern atoms.** An atom is a tuple $(p, a_1, \ldots, a_n)$; variables are terms prefixed by `?`, such as `?x`. Negated predicates use an explicit marker.
- **Horn rules.** A rule is a pair (B, H), where B is a tuple of body atoms and H is the head atom. Its semantics are $\bigwedge_{b \in B} b \rightarrow H$.
- **FOL formulas and clauses.** First-order formulas are represented as abstract syntax trees. Ground CNF clauses are stored as sets of signed literals for refutation-based satisfiability checking.
- **Causal mechanisms.** CounterBench mechanisms are represented as Boolean event dependencies, with explicit observations and intervention variables.

2. Classical Reasoning Algorithms

The notebook contains the reasoning engine, public-dataset adapters, evaluation routines, and submission-file generation in one executable artifact.

2.1 Unification and Forward Chaining

For structured knowledge bases and RuleTaker examples, unification finds a consistent substitution between rule-body patterns and known ground facts. Predicate indexing narrows candidate matches. Forward chaining repeatedly fires applicable rules until no new facts are derived or the query is established. Derived facts are held in a set, preventing duplicate derivations and allowing cyclic rules to reach a fixed point. Where enabled by the input representation, positive rules are saturated before negated body literals are considered under a closed-world interpretation.

2.2 FOL Grounding, CNF, and DPLL

For FOLIO, a recursive-descent formula parser consumes the provided `premises-FOL` and `conclusion-FOL` annotations. Quantified formulas are grounded over the finite active-constant domain; existential quantifiers introduce witness constants. The solver negates the conclusion and tests $KB \wedge \neg Q$ for unsatisfiability. Formulas are converted to negation normal form and conjunctive normal form, then passed to a recursive DPLL solver using unit propagation, pure-literal elimination, and shortest-clause branching. Unsatisfiability establishes entailment.

2.3 Counterfactual Causal Simulation

For CounterBench, the adapter constructs a Boolean causal model and answers counterfactual questions through three stages:

1. **Abduction:** enumerate exogenous-root assignments consistent with the observations and baseline model.
2. **Intervention:** apply the requested $do(X = x)$ operation, overriding the intervened variable's structural equation.
3. **Prediction:** re-simulate downstream mechanisms and evaluate the target event in the counterfactual worlds.

3. Dataset Evaluation

3.1 Sources and Evaluation Protocol

The reported run used a limit of 50 records per requested split or JSON file. RuleTaker records were downloaded from https://huggingface.co/datasets/jise/ruletaker; the evaluation result below is specifically for its `ood_test` split (50 records). The train and test splits were also downloaded, but are not included in the reported development score. FOLIO validation used 50 records from the official Yale–LILY GitHub validation mirror because the requested Hugging Face repository (https://huggingface.co/datasets/yale-nlp/FOLIO) was not directly available in this run. CounterBench records came from https://huggingface.co/datasets/CounterBench/CounterBench; 50 records were used from each of `data_balanced_alpha_V1.json` and `data_balanced_backdoor_V2.json`; the alpha sample comprised 40 Basic and 10 Conditional records, and the second file supplied 50 Backdoor records.

Accuracy is computed over supported/scored records, while coverage is the fraction of all records that the engine supports. To make unsupported cases visible in the headline result, strict accuracy is also reported as correct predictions divided by all records. The run produced 200 records in total: 196 scored, 194 correct, and 4 unsupported.

3.2 Results and Interpretation

The combined scored accuracy was $194/196 = 98.98\%$, with 98.00% coverage. The strict accuracy was $194/200 = 97.00\%$. The engine processed the 200-record evaluation in 0.045943 seconds, corresponding to approximately 0.2297 ms per input record under the reported run environment. CounterBench scored 100/100 across the sampled Basic, Conditional, and Backdoor subsets. RuleTaker OOD scored 50/50. FOLIO

Table 1: Measured results from the user's 50-record-per-split/file run. Scored accuracy is correct/scored; strict accuracy is correct/total and therefore treats unsupported records as not correct.

Evaluation subset	Total	Correct / scored	Scored acc.	Coverage	Strict acc.	Time (s)
RuleTaker OOD test	50	50 / 50	100.00%	100.00%	100.00%	0.013760
FOLIO validation	50	44 / 46	95.65%	92.00%	88.00%	0.021931
CounterBench Basic	40	40 / 40	100.00%	100.00%	100.00%	0.004435
CounterBench Conditional	10	10 / 10	100.00%	100.00%	100.00%	0.001186
CounterBench Backdoor	50	50 / 50	100.00%	100.00%	100.00%	0.004631
Overall	**200**	**194 / 196**	**98.98%**	**98.00%**	**97.00%**	**0.045943**

scored 44/46 supported cases; four of its 50 records were unsupported, giving 92% coverage and 88% strict accuracy.

The FOLIO figure is a **binary entailment metric** under the notebook's label mapping; it should not be interpreted as native three-way FOLIO accuracy. In particular, the original Uncertain category is mapped into the non-entailed class for this binary task. The four unsupported FOLIO records are excluded from scored accuracy but are reflected in coverage and strict accuracy.

3.3 Scope and Limitations

These are results on capped subsets, not full-dataset benchmark scores. In particular, the RuleTaker result is based on 50 OOD-test records, and the CounterBench result is based on 100 sampled rows across the three reported categories. The near-ceiling scores should therefore be interpreted with the sample sizes and coverage shown in Table 1. The reported inference time is specific to this execution environment and sample; it is not a hardware-independent latency guarantee. A broader evaluation should increase the per-split limit and preserve a held-out test protocol.

4. Illustrative Reasoning Traces

RuleTaker / forward chaining. Given facts $Young(Bob)$ and $Kind(Bob)$, with rules $Young(x) \rightarrow Smart(x)$ and $Smart(x) \wedge Kind(x) \rightarrow Successful(x)$, unification first derives $Smart(Bob)$ and then derives $Successful(Bob)$. The query is therefore entailed.

FOLIO / refutation. From $\forall x(Human(x) \rightarrow Mortal(x))$ and $Human(s)$, test the query $Mortal(s)$ by adding $\neg Mortal(s)$. Grounding yields $\neg Human(s) \vee Mortal(s)$, $Human(s)$, and $\neg Mortal(s)$. Unit propagation reaches a contradiction, so the negated knowledge base is unsatisfiable and the query is entailed.

CounterBench / intervention. For mechanisms $A \rightarrow B$ and $B \rightarrow C$, set the counterfactual intervention $do(A = 1)$. The simulator propagates $B = 1$ and then $C = 1$; the target event is predicted to occur in the intervened world.

Submission interface. The notebook is intended to run from top to bottom. It downloads the configured public datasets, evaluates the reasoning engine, prints accuracy, coverage, unsupported count, and inference time, and writes an `id,label` CSV for the selected test input. For a private evaluation, the same notebook can be pointed to a teacher-provided dataset in one of the supported structured formats.

5. Error Profile and Reproducibility

The aggregate score should be read together with the support rate. All 150 RuleTaker and CounterBench records were supported and classified correctly in this run. In FOLIO, 46 of 50 records were supported: 44 were correct and two were incorrect; the remaining four were unsupported. Thus, the overall score is not simply a mean of three dataset percentages: it is the micro-aggregate over the 196 supported records. Strict accuracy further penalizes the four unsupported inputs and is 97.00%.

The runtime profile is small for this sample. FOLIO accounts for 0.021931 seconds, RuleTaker OOD for 0.013760 seconds, and the two CounterBench files together for 0.010252 seconds. These measurements include the reasoning stage as timed by the notebook; dataset download time is not included in the reported inference time. The average of 0.2297 ms per input is calculated over all 200 input records, including unsupported records.

For reproducibility, the notebook configuration should retain the stated per-split/file limit of 50 and the same split selection. In particular, the reported RuleTaker score must remain tied to `ood_test`; using `train` instead would answer a different evaluation question. The RuleTaker train and test downloads are retained for the notebook's other workflows, but their records are not part of the 200-record score in Table 1. The FOLIO fallback source should also be disclosed because the measured validation records came from the Yale–LILY mirror rather than a direct Hugging Face download.

The implementation is deterministic and does not fit model parameters from these examples. Consequently, conventional parameter overfitting is not the primary concern. The relevant evaluation risks are limited sample size, benchmark-specific adapter assumptions, label conversion, and unsupported-record exclusion from scored accuracy. Reporting coverage and strict accuracy alongside scored accuracy makes these limitations explicit. A stronger follow-up evaluation would run the complete official splits, retain the OOD test protocol, and report the native three-way FOLIO labels separately if three-class classification is required.